California State University Long Beach

Mechanical and Aerospace Engineering Department



MAE 300: Engineering Instrumentation and Measurement

Experiment No. 1

Use of descriptive statistics in Physical Measurement

Lab section 19

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Abstract:

The purpose of this experiment was to apply statistical analysis to the measurement of metal washers to gain a better understanding of the ways in which this procedure can be useful within the context of engineering projects. In this experiment, students were given 15 metal washers of the same size to conduct repeated measurements using a digital caliper to record a total of 30 data measurements in millimeters (mm) to the 1st decimal place. The students would need to collaborate with other student groups to have enough variation with the metal washers that were measured. To collect enough data for the statistical analysis calculations, the students collected another 30 measurements of 10 new washers from the other student group. The mean value of the population data was 25.0 mm with a standard deviation of 0.1343. For the second part of the analysis, five data points were randomly chosen to represent a sample of the population. The sample average was the same as the population's average of 25.0 mm with a slightly lower standard deviation of 0.100, indicating less variation in the range of values for the sample data set compared to the population data set.

Introduction:

Engineers handle projects that encompass every aspect of our daily lives, from sending a text to a friend to sending an astronaut into space. There is no doubt about the scope, complexity and necessity of engineering within our world. When dealing with such high-stake projects, it is crucial that the measurements and calculations are conducted accurately, precisely, and analyzed properly. This accuracy and precision are not innate as there are many different procedures and steps necessary to ensure the best measurements and results. Utilizing statistical analysis is an invaluable process that allows engineers to analyze different volumes of data to identify trends pertaining to the data. For this reason, this experiment serves as an example of statistical analysis to provide a clear understanding of the significance of statistical analysis calculations and processes.

Background & Theory:

A digital caliper was used to collect measurements of 25 of the same type of washer. This measurement data would be recorded to the first decimal place in millimeters to provide a more precise measurement. A total of 60 measurements were required to gather enough data to create a population sized data set, which is any data set with over 30 data points. In addition to the population data set, a randomized selection of five population measurements would be used as a sample data set. Through statistical analysis of these two data sets, a comparison can be made between multiple different calculations to indicate if the sample size of five different data points is adequate to represent the whole of the population.

The calculations done for this comparison were the calculation of the mean, or average of the data using equation (1). The calculation of the median, which is the value that falls in the middle of the data set and the mode, which is the most frequent value, as well as three different

types of deviation from the calculated mean value. The average deviation from the mean was calculated with equation (2), the Root Mean Square (RMS) deviation from the mean was calculated using equation (3), and the standard deviation from the mean was calculated with equation (4). These values were calculated for both the sample and population data set and if the percentage difference between them was below 5%, it would indicate accurate representation of the population data from the sample.

In addition, a calculation of the most probable error of a single measurement (5) as well as the most probable error of the mean (6) can be calculated to see how close the measurement values and calculated mean of the population data are to the true values of the measurement and mean. A normalized histogram can also be used to see the variance of the measured values from the mean. The relative frequency of the data points that fall between the calculated class subintervals can be plotted against the class mark values to represent the distribution of measurement values among the population data set. A histogram allows for observation of the distribution curve's shape as well as if it is symmetrical or skewed in a certain direction. The Z-score and corresponding P(z) values can be calculated for each of the measurements to create a normalized distribution curve. To calculate P(z), equation (7) is used while the Z-score is found using (8). There is also a table that can be used to find the corresponding P(z) value if given a certain Z-score. If the mean and median are equal, then the distribution of the normalized distribution curve is likely to be symmetrical.

Graphical representations like the normalized histogram and normalized distribution curve help identify outliers and potential sources of error within the experimental procedure. When measuring identical items, increasing the sample size usually results in a distribution that more closely resembles a bell-shaped curve normal distribution curve. In contrast, smaller samples may show greater variability and may not accurately represent the population. These statistical results can be visualized by using histograms and other graphical plots.

(1): $\mu = \frac{\sum x_i}{n}$.

(2): $RMS = \frac{\Sigma|x_i - \bar{X}|}{n}$,

(3): $avg\ deviation = \sqrt{\frac{\Sigma(x_i - \bar{X})^2}{n}}$

(4): $s = \sqrt{\frac{\Sigma(x_i - \bar{X})^2}{n-1}}$

(5): $E_s = 0.6745 * \sqrt{\frac{\Sigma(x_i - \bar{X})^2}{n-1}}$

(6): $E_m = \frac{E_s}{\sqrt{n}}$

(7): $P(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/e}$

(8): $Z = \frac{x_i - \mu}{\sigma_x}$

Experimental Procedures:

The experiment began with the student groups selecting a random set of 15 metal washers from the container provided by the instructor. Each group was also given the same type of digital caliper and told to collect the measurement washers in mm to the 1st decimal place. The groups collected a total of 60 measurements of different washers of the same size and recorded them within a Microsoft Excel spreadsheet. This data would then be used to conduct statistical analysis of the population data set, all 60 of the measurements, and another set of sample data of five randomly selected measurements. These analyses and calculations of the population and sample data would be compared to draw comparisons of the two sets of data.

Raw data:

Table 1.1: Initial Measurements of 15 Washers (mm)

25.1	24.9	25.0	25.1	25.0	24.8
25.0	25.1	24.8	24.8	25.1	24.9
24.7	25.1	25.0	25.1	25.1	25.0
25.1	24.9	25.0	25.0	24.8	25.1
25.1	24.8	25.0	24.9	24.7	25.0

Table 1.2: Additional Measurements of 10 Washers (mm)

24.7	24.9	24.8	25.1	24.9	25.0
24.9	25.0	25.1	25.1	25.1	25.1
25.0	25.1	24.6	25.1	24.9	24.9
25.1	24.8	25.1	25.0	25.1	24.9
25.1	25.1	25.1	25.1	24.8	24.8

Table 1.3: Selected Sample Data (mm)

Sample Data (mm):	25.1	24.9	25.1	24.9	25.0

Sample Calculations and Results:

$$E_s = 0.6745 * \sqrt{\frac{\Sigma(x_i - \bar{X})^2}{n-1}} = 0.6745 * \sqrt{\frac{1.08183}{60-1}} = 0.09133$$

$$E_m = \frac{E_s}{\sqrt{n}} = \frac{0.09133}{\sqrt{60}} = 0.0118$$

$$Average\ Deviation = \frac{\Sigma|x_i - \bar{X}|}{n}$$

$$Root\ Mean\ Square\ (RMS) = \sqrt{\frac{\Sigma(x_i-\bar{X})^2}{n}}$$

$$Standard\ Deviation\ (s) = \sqrt{\frac{\Sigma(x_i-\bar{X})^2}{n-1}}$$

Table 1.4: Sample Data Calculations for Standard Deviation (s)

Diameter (x_i) (mm)	$x_i - \bar{X}$	$(x_i - \bar{X})^2$
25.1	0.1	0.01
24.9	-0.1	0.01
25.1	0.1	0.01
24.9	-0.1	0.01
25.0	0.0	0.00

Relative Frequency & Histogram Calculations:

$X_{min} = 24.6$ $X_{max} = 25.1$

$\Delta X = \frac{(X_{max}-X_{min})*1.2}{\#\ of\ Class\ Intervals} = 0.1$

$X_1 = X_{min} - \frac{\Delta X}{2} = 24.56$

$X_2 = X_1 + \Delta X = 24.65$

$X_3 = X_2 + \Delta X = 24.75$

$X_4 = X_3 + \Delta X = 24.85$

$X_5 = X_6 + \Delta X = 24.95$

$X_6 = X_5 + \Delta X = 25.05$

$X_7 = X_6 + \Delta X = 25.15$

$X_8 = X_7 + \Delta X = 25.25$

$$Class\ Marks = \frac{X_{start} + X_{end}}{2}$$

$$Relative\ Frequency = \frac{Frequency}{Total\ number\ of\ Values}$$

Table 1.4: Z-Score Calculation Values

Class	Class Subintervals		Class Marks	Frequency	Relative Frequency
	Start	End			
1	24.56	24.65	24.6	1	0.02
2	24.65	24.75	24.7	3	0.05
3	24.75	24.85	24.8	9	0.15
4	24.85	24.95	24.9	10	0.17
5	24.95	25.05	25.0	13	0.22
6	25.05	25.15	25.1	24	0.40
7	25.15	25.25	25.2	0	0.00

Figure 1.1: Normalized Histogram of Population Data

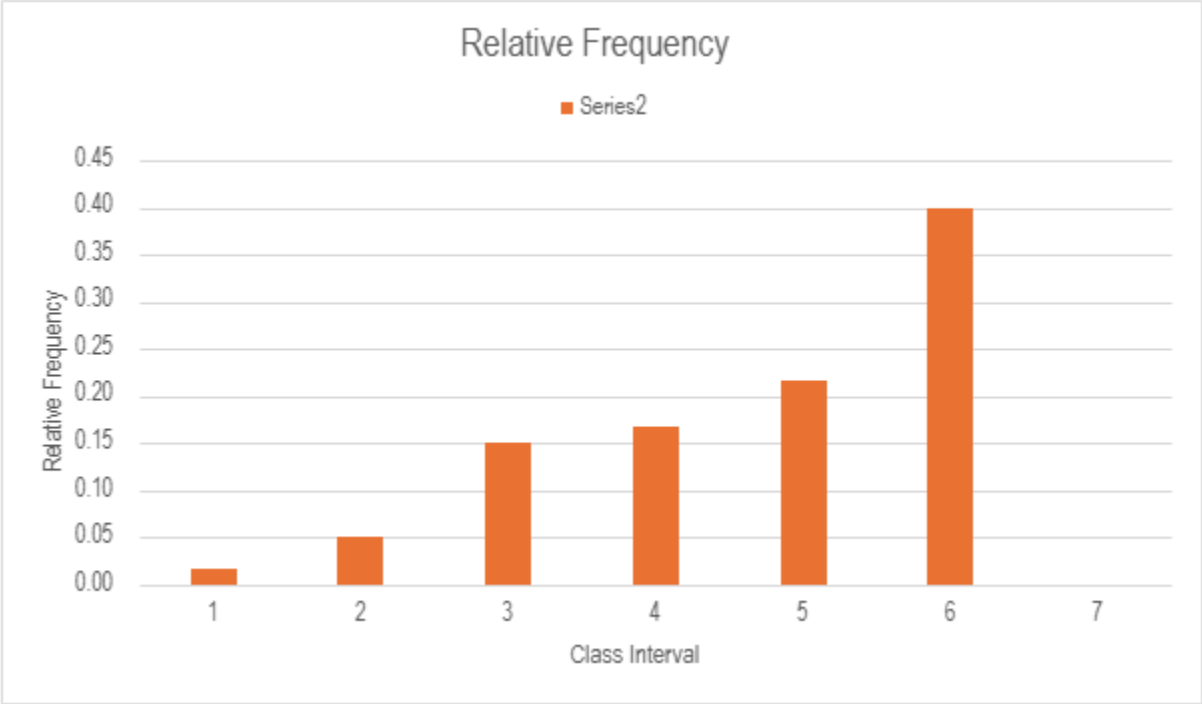


Table 1.5: Z and P(z) Values for Population Data

X	Z	P(z)
25.1	0.96	0.25
24.9	-0.53	0.35
25.0	0.21	0.39
25.1	0.96	0.25
25.0	0.21	0.39
24.8	-1.28	0.18
25.0	0.21	0.39

25.1	0.96	0.25
24.8	-1.28	0.18
24.8	-1.28	0.18
25.1	0.96	0.25
24.9	-0.53	0.35
24.7	-2.02	0.05
25.1	0.96	0.25
25.0	0.21	0.39
25.1	0.96	0.25
25.1	0.96	0.25
25.0	0.21	0.39
25.1	0.96	0.25
24.9	-0.53	0.35
25.0	0.21	0.39
25.0	0.21	0.39
24.8	-1.28	0.18
25.1	0.96	0.25
25.1	0.96	0.25
24.8	-1.28	0.18
25.0	0.21	0.39
24.9	-0.53	0.35
24.7	-2.02	0.05
25.0	0.21	0.39
24.7	-2.02	0.05
24.9	-0.53	0.35
24.8	-1.28	0.18
25.1	0.96	0.25
24.9	-0.53	0.35
25.0	0.21	0.39
24.9	-0.53	0.35
25.0	0.21	0.39
25.1	0.96	0.25
25.1	0.96	0.25
25.1	0.96	0.25
25.1	0.96	0.25
25.0	0.21	0.39
25.1	0.96	0.25
24.6	-2.77	0.01
25.1	0.96	0.25
24.9	-0.53	0.35
24.9	-0.53	0.35
25.1	0.96	0.25
24.8	-1.28	0.18

25.1	0.96	0.25
25.0	0.21	0.39
25.1	0.96	0.25
24.9	-0.53	0.35
25.1	0.96	0.25
25.1	0.96	0.25
25.1	0.96	0.25
25.1	0.96	0.25
24.8	-1.28	0.18
24.8	-1.28	0.18

Figure 1.2: Z vs. P(z) Standard Normal Distribution Curve

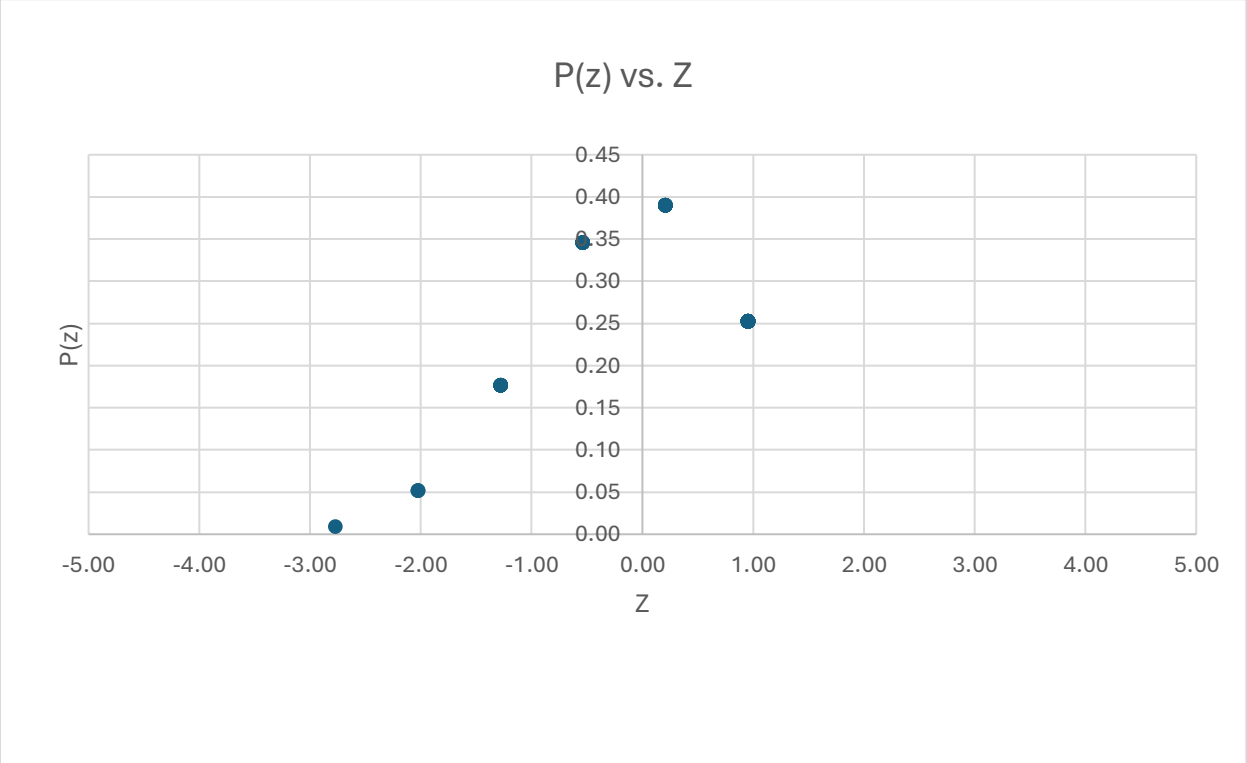


Table 1.6: Comparison of Sample and Population Calculations

Calculated Results	Sample	Population	% Difference
Range	0.20	0.50	60.0%
Midrange	25.00	24.85	3.3%
Mean	25.00	24.97	0.12%
Mode	25.10	25.10	0.0%
Median	25.00	25.00	0.0%
Average Deviation from Mean	0.08	0.11	27.3%
RMS Deviation from Mean	0.09	0.13	30.8%
Standard Deviation from Mean	0.10	0.13	23.1%

Discussion of results:

Part 1:

The first part of the experimental calculations required the calculation of the most probable error of a single measurement (E_s) and the most probable error of the mean (E_m). E_s is the value that represents the range in which there is a 50% probability of the true measurement value falling within. E_s is calculated as 67.45% of the standard deviation value, which was 24.97 mm for our population data. The E_s was ± 0.09133, this value is relatively small which means that there is a narrow range of values that are likely to contain the true value of the washer's measurement. In addition, the calculated E_m, which represents the range in which the true value of the mean is expected to fall within. This value was smaller than the calculated E_s as the $E_m = \pm 0.0118$.

The second part of the experimental calculations called for the creation of a normalized histogram as well as a standardized normal distribution curve. The normalized histogram showcased the relative frequency of certain values within the population data set. As we had a total of 60 measurements, this meant that the resulting histogram would need seven class intervals or ranges between start and end class subinterval values. These subinterval values were calculated from the minimum value of 24.6 mm and a maximum value of 25.1 mm. There was a greater concentration of measurement values that fell into the larger class subinterval ranges, in particular the largest volume of values fell into the sixth-class interval, meaning that there was the highest frequency of measurements that fell within that subinterval range, 25.05 mm to 25.15 mm. This histogram was not symmetrical as it was more rightly skewed to the larger values, which makes sense when looking at the mode as being 25.1 mm for the sample and population data sets.

The normal distribution curve was plotted with the Z-scores and correlated P(z) values. The Z-score represents the number of standard deviations above or below the mean that a single measurement value falls into. The largest Z-score that was calculated came from the 24.6 mm measurement, with a Z-score of -2.77, which means that it almost falls into three standard deviations below the mean. Overall, the range of Z-score values fell within the range of 0.96 to -1.28, meaning that most of the measurements were one standard deviation from the mean or less. The two main outliers within our normalized distribution curve were the Z-score values of -2.77 and -2.02, indicating that our population data outliers were below the mean value of 24.97 mm.

After analyzing the normalized histogram and standard normal distribution curve from the calculations of the measurements, there is a way to identify potential sources of error within the experimental procedure. Although the shape of the standardized normal distribution curve follows Chebyshev's Theorem for Normal Distribution by showcasing a bell-curved shape, there is a clear negative skew to the data. This means that out of the collected measurements within the population data set, there were more values that were larger than the mean. This is also reflected in the mode value, 25.1 mm, being greater than the mean value of 25.0 mm. This leads to a

potential source of error with the individuals instructed to measure the washers, as they would have tended to record larger values.

Part 2:

When comparing the calculated values of the mean, median, and mode of the sample and population data, there is little to no difference in these values. The identical mode values are due to the high frequency of 25.1 mm as a measurement, as it was most likely to be pulled into the randomly selected sample data set. In addition, the high frequency of the 25.0 mm measurement would explain the similarities in the mean and median values as this distribution would have been reflected in the random sample. However, when analyzing the percentage difference of the range, a sample of 5 data points is not an accurate representation of the entire population data. The range of values for the sample and population data had a large percentage difference of 60.0%. When referring to the engineering standard of 5% difference as being an indication of a good agreement between data, the sample and population data sets do not align in this regard. In addition, the Average Deviation, RMS, and Standard Deviation values of the data sets had a percentage difference of 27.3%, 30.8%, and 23.1% respectively. These high percentage differences can be attributed to the larger range of values within the population data, as this would increase the values for the calculated deviations from the mean. Therefore, the sample size of 5 data points is insufficient to represent the entire population data set.

The bias error for this experiment was reflected in the P(z) vs z graph, as its standard normal distribution curve was skewed to the left. While getting the measurements, the person handling the caliper may have developed a habit of holding the washer at a certain angle to get the measurements as quickly as possible. This is one possible error that may have led to the shewed graph. To minimize this error in and improve the results, the team conducting the experiment could rotate with each other to get the measurements.

Another of the common kinds of errors that may have contributed to the shewed graph is the environmental and material constraint. Due to the heat transfer from the hand holding the washer conducting the diameter measurement, the diameter may have slightly changed. To minimize this error, the team could have worn gloves to avoid contact with the experimental washers.

Conclusions and Recommendations:

This experiment required students to collect 60 measurements of metal washers using a digital caliper and to conduct statistical analysis on the collected values. The 60 measurements would represent the population data set while a group of 5 randomly selected measurements would represent a sample data set of the population. Subsequent calculations of the mean, median, mode, range, and deviations from the mean would be calculated for both data sets and draw conclusions about the accuracy of the sample as a representation of the population data. In addition, the creation of a normalized histogram and normalized distribution curve would be created for the population data to identify any outliers within the population data set.

The calculation of the standard deviation and mean value from the population data allow the conclusion that the metal washer is 24.97 $\pm$0.13 mm in diameter. The normal distribution curve reveals the outlier measurement values as 24.6 mm and 24.7 mm, which are both below the mean calculated value of 24.97 mm.

Comparing the average, RMS, and standard deviation from the mean values of the sample and population data resulted in a percentage difference much greater than the engineering standard of 5% percentage difference for a good agreement between the data sets which means that the sample of 5 randomly selected measurements are not a good representation of the entire population. In conclusion, a statistical analysis of our measurement values revealed a potential source of error within our measurement process. The values that were calculated within this experiment showcase how the values that were recorded tended to be larger than the calculated average of the measured values, with the outliers falling below the calculated mean.

Bibliography and References:

Wang, Tom. " Complete Lectures and Experiments File ." pdf.*slides*, 27 Jan 2026, https://csulb.instructure.com/courses/116603/files/28679338/download?download_frd=1